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Adolescent and Parent Experiences and Perceived Effectiveness of Physical Interventions for Chronic Musculoskeletal Pain: A Qualitative Study

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Received: 24 September 2024 | **Revised:** 7 October 2024 | **Accepted:** 10 October 2024

Funding: The authors received no specific funding for this work.

Keywords: adolescent | child | chronic pain | musculoskeletal pain | parent | physical therapy | qualitative

ABSTRACT

Introduction: There is a paucity of evidence on the effectiveness of physical interventions for chronic musculoskeletal pain affecting adolescents. This study explored adolescent and parent experiences and perceived effectiveness of physical intervention for chronic musculoskeletal pain in the UK.

Methods: Using semi-structured interviews, 21 young people receiving treatment for chronic musculoskeletal pain and 21 parents were interviewed regarding their healthcare experiences. Data regarding experiences of physical interventions were analysed in this secondary qualitative analysis using inductive reflexive thematic analysis.

Results: Three themes that captured why adolescents and parents perceived physical interventions to be effective or not were (1) feeling in control, (2) the right person and an explanation that makes sense and (3) make a difference to me not my foot.

Discussion: Core concepts that appeared to influence the perceived effectiveness of physical interventions from the perspectives of adolescents and parents included facilitating adolescent autonomy, establishing a therapeutic alliance on the basis of trust and focusing physiotherapy treatment on meaningful participation in life.

Conclusion: There were no ideal physical interventions. The themes convey the importance of therapeutic alliance, intervention context and the need for personalised care.

1 | Background

Musculoskeletal pain is the most prevalent chronic pain location (25.7%) affecting children and adolescents (Chambers et al. 2024). It can continue into adulthood (Leino-Arjas et al. 2018; Rathleff et al. 2016) and poses a major health concern (King et al. 2011). Conditions such as low back pain are a leading cause of global disability (GBD 2021 Diseases and Injuries Collaborators 2024; GBD 2019 Diseases and Injuries Collaborators, 2020). There are substantial gaps in understanding musculoskeletal pain affecting adolescents (Kamper and Williams 2017) along with underprioritisation and lack of investment within services and research (Eccleston et al. 2021). This is important to address because in a cross-sectional child and adolescent musculoskeletal pain cohort study (n = 100), adolescents affected by musculoskeletal pain experience physical, social and emotional impacts with most (62%) not expecting a pain-free future (Pourbordbari et al. 2022).

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Physiotherapists (also known as physical therapists) routinely promote physical interventions and support self-management in adolescents experiencing musculoskeletal pain. Physical interventions involve 'active' interventions such as body movement or energy expenditure (e.g., muscle strengthening and functional activities) and 'passive' interventions (e.g., massage and taping). Globally, physical interventions are recommended as part of the World Health Organization guidelines on the management of chronic pain in children; however, this is based on a low certainty of evidence (World Health Organization 2020a, 2020b). Cochrane (Nascimento Leite et al. 2023) and systematic reviews (Lyng et al. 2023) have identified a paucity of evidence into the effectiveness of physical interventions for adolescents who experience chronic noninflammatory pain with evidence focused on inflammatory conditions such as juvenile idiopathic arthritis.

Furthermore, qualitative research has identified that physical interventions for young people with chronic pain can be associated with worsening pain intensity, child and family boredom, poor adherence and family conflict (World Health Organization 2020b); however, in clinical trials, the reporting of possible side effects has not been adequately reported (Lyng et al. 2023). Within their call for action, Eccleston et al. (2021) identify there is a need for more rigorous evaluation of physical interventions to identify the role of specific techniques and to explore the influence of the child and parent relationship with the therapist. To embrace this complexity, physical interventions need to be explored in real-world settings.

Addressing a significant knowledge gap, this study sought to explore adolescent and parent experiences and perceived effectiveness of physical interventions for chronic musculoskeletal pain in the United Kingdom. Developing an understanding of the experiences and perceived effectiveness of physical interventions from a user perspective will provide important knowledge that will assist clinical decision-making and service delivery and inform future clinical trials.

2 | Method

2.1 | Study Design

This study comprised a secondary qualitative analysis of data from a previous programme of research informed by Q-methodology (Brown 1993, 1996; Stephenson 1993) and led by the main author (Joslin et al. 2021, 2023). Secondary analysis, a known methodology, is most frequently used when one or more authors have been involved in the primary research (Heaton 2008). The primary research explored the experiences and views of adolescents and parents regarding the outcomes that were important to measure during chronic musculoskeletal pain treatment. Although the primary study explored topics related to the treatment experience, only data that addressed the original research question were analysed. Data in relation to the experience of physical interventions did not meet the scope of the study and are analysed in this secondary analysis study. The secondary qualitative analysis used the full sample of 21 adolescents and 21 parents as described in the primary study because all participants described an experience of physical interventions as part of their multidisciplinary treatment.

2.2 | Ethical Approval

Ethical HRA approval was granted on 30 April 2018 (IRAS ID: 230317 REC Reference number: 18/SC/0138). Written consent was provided by participants for further secondary analysis under the supervision of the lead author.

2.3 | Participants

Participants were purposively sampled for sex (child and parent), adolescent age and treatment stage. The aim was to recruit between 18 and 24 young people and 18 and 24 parents. This was within the broad target sample size recommended for qualitative interviews (Adler and Adler 2012). Participants were recruited between May 2018 and April 2019 from local services (i.e., physiotherapy departments, rheumatology clinics and child psychology services) and tertiary multidisciplinary chronic pain services at two National Health Service (NHS) hospital sites in England. Adolescents were eligible for inclusion in the study if they were (1) aged between 11 and 18 years old, (2) receiving treatment for chronic musculoskeletal pain in NHS services or had been discharged less than 1 month ago, (3) able to communicate in English without assistance and (4) able to provide written consent. Parents were eligible if their child fulfiled the above criteria, could communicate in English and provided consent. Chronic musculoskeletal pain was defined as pain located over the muscle, joints or soft tissue; lasting more than 3 months and where systemic disease (e.g., juvenile idiopathic arthritis) had been medically excluded.

2.4 | Recruitment

Initially, clinical staff mailed the study information to all eligible participants. Staff then provided eligible participants with the study information during face-to-face appointments. If interested, parents contacted the lead author directly. Over the recruitment period, the lead author attended specific clinics (in a separate room). This meant that 9 parents and 10 young people had chosen to meet the researcher prior to the interview. Participants had no prior relationship with the research team. A young person could choose to participate and their parent could decline, a parent could participate and their child could decline or they could both participate. Both parents of the same child could also participate. All participants received the study information over 48 h prior to providing written consent to participate. Following the Research Ethics Committee guidance, for this study, it was felt that parents should be aware of their child's involvement, but adolescents could give their own consent and agree to be an active participant without coercion. Participants did not receive any payment for participating and did so voluntarily.

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2.5 | Data Collection

Data were collected during a single semi-structured interview, where participants drew a timeline of their treatment and experiences (Hurtubise and Joslin 2023). The lead author interviewed all participants and was introduced as a researcher. They were concurrently working as a specialist physiotherapist at a different hospital site and were not involved in any of the participants' treatment. Participants chose to be interviewed at home, in hospital, via the telephone or using online methods (video call or messaging platform). The single semi-structured interview schedule had open questions and probes that explored important changes (positive and negative) as a response to multidisciplinary treatment that included physical interventions. The interview schedule (Supporting Information S1) was adapted with two patient representatives using instructions published by Save the Children Norway (2008). All interviews were audio-recorded using a digital voice recorder (Olympus WS-852) and transcribed verbatim. The transcripts were not returned to participants because the timeline served as a written summary of what was discussed and participants could amend this as they wished throughout the interview. Pseudonyms were used and any information that could potentially identify participants was removed. Field notes were made immediately after the interview, and all participants received a copy of the final results. This manuscript was prepared in accordance with the consolidated criteria for reporting qualitative research guidelines (Tong, Sainsbury, and Craig 2007).

2.6 | Data Analysis

Themes and patterns across the dataset were analysed using inductive reflexive thematic analysis (Braun and Clarke 2006, 2019, 2022). An experimental approach was used to capture and explore adolescents and parents' perspectives and experiences and how they made sense of their reality. Analysis was semantic, with meaning explored at an explicit level and effectiveness of physical interventions established by participant report of change (positive or negative) or no change. The theoretical paradigm of critical realism allowed for the exploration of lived experience in a 'real world' context (Lawani 2021). Reflective thematic analysis followed the six stages outlined by Braun and Clarke (2006, 2022): (1) data familiarisation; (2) systematic data coding; (3) generating themes; (4) review themes; (5) defining, refining and naming themes and (6) report writing. Data analysis was completed without the use of software. The transcripts were coded line by line and all quotes related to physical interventions were printed, highlighted and separated so that they could be moved into groups and common threads. The two authors discussed and finalised data coding. During a 2-h face-to-face meeting, the two authors generated themes by using mind maps and then rearranged codes to represent key threads on a whiteboard. The themes were then revisited as they wrote and edited the report. Credibility was established through detailed discussions between the authors, documentation of the decision trail, data triangulation using different methods (interview and timelines) and the inclusion of quotes across the sample in the results (Noble and Smith 2015). As this was secondary analysis, the decision to stop data collection was established in the primary study when no new codes were established in the final three interviews.

3 | Results

Forty-two interviews (21 adolescents and 21 parents) provided an understanding of 24 different treatment narratives. The narratives were reported from different viewpoints: 6 narratives were only from the adolescent perspective, 3 narratives were only from the parent perspective and 12 narratives were from multiple perspectives including the adolescent and 1 parent (n = 10) or the adolescent and both parents (n = 2). One adolescent and parent chose to be interviewed together. For the remaining 11 narratives, the parent(s) and young person were interviewed separately. Table 1 outlines the sex and age of the young person in each of the 24 narratives alongside the location of musculoskeletal pain. The duration of pain reported by participants ranged from 5 to 120 months. The experiences of parents included 4 fathers and 17 mothers. The length of the interview ranged from 28 to 98 min. The quote numbers in the results represent the participant number and do not align with the narrative numbers to avoid deductive disclosure.

3.1 | Type and Delivery of Physical Interventions

Table 2 outlines the range of active and passive physical interventions described by participants. The aim of each intervention was mapped to the International Classification of Functioning in terms of body structure and function, activity and participation (World Health Organization 2001). Most physical interventions were delivered by physiotherapists in an outpatient setting; therefore, when referring to health professionals from this point, the term physiotherapist will be used.

3.2 | Themes

Reflexive thematic analysis established three themes that were key to understanding how an adolescent and their parent experienced physical interventions for musculoskeletal pain and how these experiences appeared to relate to perceived effectiveness of physical interventions: (1) feeling in control, (2) the right person and explanation that makes sense and (3) make a difference to me not my foot.

3.2.1 | Theme 1: Feeling in Control

This theme captures the importance of parents and physiotherapists giving adolescents choices and autonomy over their treatment and future. It portrays the changing dynamic of autonomy during adolescence and the innate protective reactions of adults (parents and physiotherapists) to solve problems for young people. When a physical intervention was reported as effective, the young person was actively involved in making the decision to participate, could stop when they chose and understood how it linked to their future.

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TABLE 1 | Musculoskeletal presentations of adolescents in each narrative.

Narrative	Reported sex of adolescent	Reported age of adolescent	Localised or widespread pain	Unilateral or bilateral	Location of pain
1	Female	14	Widespread	Bilateral	Low back, shoulder, elbow, hip and knee
2	Female	18	Widespread	Bilateral	Low back, thoracic, chest (rib), wrist and ankle
3	Female	15	Localised	Unilateral	Right foot
4	Male	12	Localised	Bilateral	Chest and low back
5	Female	16	Localised	Unilateral	Right foot
6	Female	14	Localised	Bilateral	Knee
7	Female	17	Widespread	Bilateral	Low back, thoracic, neck shoulder, wrist and ankle
8	Male	14	Widespread	Bilateral	Low back, neck, jaw, shoulder, knee and ankle
9	Female	15	Localised	Unilateral	Right knee
10	Female	15	Localised	Unilateral	Left foot
11	Female	15	Localised	Unilateral	Right foot
12	Female	13	Widespread	Bilateral	Knee, ankle, shoulder, elbow, wrist and hand
13	Female	15	Localised	Unilateral	Left sided chest
14	Male	11	Localised	Unilateral	Right heel pain
15	Female	15	Widespread	Bilateral	All of the spine and trunk, all of the upper limbs and all of the lower limbs
16	Male	12	Localised	Unilateral	Left hand pain
17	Male	11	Widespread	Bilateral	All of the lower limbs
18	Female	15	Localised	Bilateral	Neck pain
19	Female	15	Widespread	Bilateral	Total body pain from head to toes
20	Female	12	Localised	Unilateral	Left foot and ankle
21	Female	14	Localised	Unilateral	Right ankle pain
22	Female	14	Localised	Unilateral	Right shoulder pain
23	Female	15	Localised	Unilateral	Left knee pain
24	Female	14	Localised	Bilateral	Ankle pain

The experiences of desensitisation exercises best portrayed the key concept of the theme. Positive effects of desensitisation exercises were described by an adolescent who performed desensitisation themselves. They were in control of starting and stopping the exercise and focused on being able to wear shoes, a goal they considered important. In contrast, negative effects of desensitisation exercises were described when adolescents had a massage, touch or water applied to the painful limb by physiotherapists or parents. The adolescent was not actively involved (often lying and watching) and the rationale was that this intervention reduced sensitivity and pain. Parents described struggling to watch their child have the intervention performed on them, and one parent recalled their child screaming 'no water' to stop the physiotherapist. Young people reported fear, increased pain and reduced function because of the intervention. In a similar way, when home exercise programmes were prescribed by physiotherapists and enforced by parents without the involvement of the young person, worsening family conflict was described. When young people voiced their concerns, their opinions felt muted:

> Adolescent 16: The exercises we were given never fit in and I'd go back [to physiotherapy] and they would say do more and I'm like, this is not going to work and there would be arguments between me and my mum and dad about when to do physio and you have to do it.

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TABLE 2 | Type and delivery of physical intervention experiences reported.

Physical intervention	Description	Active or passive	International classification of functioning (World Health Organization 2001) the intervention aims to change	Number of the 24 narratives that described the intervention (number of times intervention reported)	Number of times intervention described as having positive effect	Number of times intervention described as having no effect	Number of times intervention described as having negative effect	Health professionals reported to deliver the intervention	Setting: Inpatient or outpatient
Provision of aids and equipment	Provision of wheelchair	Passive	Activity	3 (3)	0	0	3	Occupational therapist ($n = 1$) unknown ($n = 2$)	Outpatient ($n = 3$)
	Provision of walking aids, for example, crutches	Passive	Activity	8 (8)	1	0	7	Physiotherapist ($n = 8$)	Outpatient ($n = 7$) Inpatient ($n = 1$)
	House adaptions for example, rails on stairs	Passive	Activity	1 (1)	1	0	0	Occupational therapist ($n = 1$)	Outpatient ($n = 1$)
Functional land-based activity (adolescent supervised by health professional)	Walking and stairs practice improving weight-bearing, independence, distance and speed	Active	Activity	7 (13)	11	0	2	Physiotherapist ($n = 13$)	Outpatient ($n = 6$) Inpatient ($n = 7$)
	Graded return to meaningful activity, new activities and sports	Active	Participation	11 (12)	12	0	0	Physiotherapist ($n = 12$)	Outpatient ($n = 10$) Inpatient ($n = 2$)
	Graded exposure	Active	Participation	4 (4)	4	0	0	Physiotherapist ($n = 2$) Occupational therapist ($n = 2$)	Outpatient ($n = 2$) Inpatient ($n = 2$)
	Coaching quality of movement	Active	Activity	1 (1)	1	0	0	Physiotherapist ($n = 1$)	Outpatient ($n = 1$)
	Balance challenges, for example, wobble boards	Active	Activity	1 (1)	1	0	0	Physiotherapist ($n = 1$)	Outpatient ($n = 1$)

(Continues)

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TABLE 2 | (Continued)

Physical intervention	Description	Active or passive	International classification of functioning (World Health Organization 2001) the intervention aims to change	Number of the 24 narratives that described the intervention (number of times intervention reported)	Number of times intervention described as having positive effect	Number of times intervention described as having no effect	Number of times intervention described as having negative effect	Health professionals reported to deliver the intervention	Setting: Inpatient or outpatient
Prescribed home exercise plan (completed independently at home)	Mixed exercises tailored to exam findings	Active	Body structure and function	8 (9)	1	3	5	Physiotherapist ($n = 9$)	Outpatient ($n = 9$)
	Stretches to increase muscle flexibility	Active	Body structure and function	4 (5)	1	3	1	Physiotherapist ($n = 4$) Chiropractor ($n = 1$)	Outpatient ($n = 5$)
	Core stability exercises	Active	Body structure and function	2 (2)	1	1	0	Physiotherapist ($n = 2$)	Outpatient ($n = 2$)
	Range of motion exercises	Active	Body structure and function	5 (5)	4	1	0	Physiotherapist ($n = 5$)	Outpatient ($n = 5$)
	Exercises using equipment, for example, gym ball	Active	Body structure and function	3 (3)	2	0	1	Physiotherapist ($n = 3$)	Outpatient ($n = 3$)
	Weight-bearing exercise	Active	Activity	1 (2)	0	0	2	Physiotherapist ($n = 2$)	Outpatient ($n = 2$)
	Self-management plan: What exercises to do when	Active	Participation	1 (1)	1	0	0	Physiotherapist ($n = 1$)	Inpatient ($n = 1$)
Group exercise/ activity	Exercise class including yoga, pilates or gym work	Active	Activity	4 (4)	0	3	1	Physiotherapist ($n = 4$)	Outpatient ($n = 4$)
Water-based exercise and activity	Hydrotherapy	Active	Activity	8 (10)	9	0	1	Physiotherapist ($n = 10$)	Outpatient ($n = 8$) Inpatient ($n = 2$)

(Continues)

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TABLE 2 | (Continued)

Physical intervention	Description	Active or passive	International classification of functioning (World Health Organization 2001) the intervention aims to change	Number of the 24 narratives that described the intervention (number of times intervention reported)	Number of times intervention described as having positive effect	Number of times intervention described as having no effect	Number of times intervention described as having negative effect	Health professionals reported to deliver the intervention	Setting: Inpatient or outpatient
Passive physical interventions	Desensitisation	Passive	Body structure and function	3 (3)	1	0	2	Physiotherapist ($n = 3$)	Inpatient ($n = 1$) Outpatient ($n = 2$)
	Use of K-tape	Passive	Body structure and function	1 (2)	2	0	0	Physiotherapist ($n = 2$)	Outpatient ($n = 2$)
	Soft tissue massage	Passive	Body structure and function	1 (1)	1	0	0	Physiotherapist ($n = 1$)	Outpatient ($n = 1$)
	Posture correction	Passive	Body structure and function	1 (2)	0	2	0	Physiotherapist ($n = 2$)	Outpatient ($n = 2$)
	Acupuncture	Passive	Body structure and function	1 (1)	Mixed[a]	—	—	Physiotherapist ($n = 1$)	Outpatient ($n = 1$)
	Insoles/foot orthotics	Passive	Body structure and function	1 (1)	1	0	0	Podiatrist ($n = 1$)	Outpatient ($n = 1$)
	Manual techniques	Passive	Body structure and function	3 (3)	1	2	0	Osteopath ($n = 2$) Chiropractor ($n = 1$)	Outpatient ($n = 3$)
Mirror therapy	Mirror therapy	Active	Body structure and function	1 (1)	0	0	1	Physiotherapist ($n = 1$)	Inpatient ($n = 1$)

[a]Mother reported that their child had reported an initial improvement after acupuncture but was unsure because their child reported it 'got bad again'. The adolescent did not discuss acupuncture.

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When young people were involved in goal setting and decision-making, they described physical interventions positively. The process of parents and physiotherapists taking a step back and empowering the young person to make their own decisions could take time. Parents reflected on initial futile searches for a quick fix for the pain, unintentionally disempowering their child. Parents recognised the importance of changing the approach and the need for physiotherapists and parents to support adolescents to make their own choices and lead their recovery.

> Mother (Parent 3): I spoke to [the physiotherapist] just this week and said '[physiotherapist], are you willing to help if she wants you to do it?' and she said 'absolutely, I'm here for you, whatever, just ring me up when she's ready'. But she [physiotherapist] knows that she can't force that and that's what been our progress, I think... her [daughter] feeling in control.

3.2.2 | Theme 2: The Right Person and Explanation That Makes Sense

This theme highlighted the importance of a therapeutic relationship and treatment starting with a mutual understanding of why the young person was experiencing pain. Adolescents and parents reported that physiotherapists needed the right skills, attributes, knowledge and behaviours to provide the optimal treatment for each individual patient. They wanted to be able to trust their physiotherapist. They talked positively about physiotherapists who conveyed optimism, demonstrated expertise in musculoskeletal pain conditions and whose communication skills meant they felt listened to. Parents described how behaviours, such as the physiotherapist providing a contact via email or telephone, offered a valuable source of support. Although skills and expertise can be developed, the personal connection was described as something that cannot be trained:

> Father (Parent 14): Physiotherapy I think works when you have a patient who's willing and a therapist who gets the patient, that relationship, and you can't manufacture that, and it worked well for hydrotherapy.

For the treatment to be considered correct, it needed to match a credible pain explanation, and this appeared to be a vital first step. Positive effects were described when physiotherapists were able to reassure young people and parents that their condition was reversible and not a sign of something serious or degenerative.

> Mother (Parent 19): I think the physio is definitely helping... just feeling acknowledged that it isn't something that is just going to get worse and worse and worse through her life, has made her feel better.

This repeated reassurance from physiotherapists that pain did not equal harm in 'real time' (while doing an exercise and/or activity) was described positively in comparison to performing exercises at home independently.

A credible explanation then needed to align with adolescent and parent expectations of the required physical intervention (access, type, intensity, timescale for change and outcome). Parents felt let down and reported negative effects when access was delayed, the intensity of treatment was not frequent enough or the intervention was not what they expected. In many cases, parents expected physiotherapists to provide passive not active interventions and wanted to see physiotherapists physically doing something with their child.

> Mother (Parent 13): ...their physio isn't physio, because they basically said which I do agree with to an extent it's about getting her back to normal, back to normal routines ...So, physio to me were [sic] weekly meetings...they told me she does have physio it's not just physio in the typical sense of physio. I was like, okay, we will have to agree to disagree on that.

3.2.3 | Theme 3: Make a Difference to Me Not My Foot

This theme highlighted that a physical intervention needed to be centred around the person not the body part that hurts. If the young person and parent perceived the intervention would have a direct effect on body structure (tissue healing, muscle strength or flexibility) or body function (pain), negative effects were reported. In contrast, if the rationale for an intervention was to increase a young person's meaningful participation in life, immediate and long-lasting positive changes were reported. Interventions that are traditionally considered to be influencing body structure and function, such as using tape for knee pain, were reported as having a positive effect when the rationale was focused on participation. For example, to give the young person confidence to play their first full game of hockey.

The provision of aids and equipment helps to illustrate how a biomedical rationale appeared to facilitate negative effects. Wheelchairs, slings and crutches are used with acute injuries to protect a specific part of the body and allow healing of body tissues. In chronic pain, there is no injury to protect and therefore, no healing time to guide the type or timing of aid provision. As a result, adolescents described being reliant on walking aids for months and in some cases for years. Young people described adverse effects of wheelchairs and crutches that included new onset of tiredness/fatigue, increase in pain locations, lower mood and social isolation and victimization. Home exercise programmes, mindfulness and mirror therapy were also described by adolescents as drawing attention to the painful body part. Physiotherapy was a reminder of their problem and the pain. Adolescents explained they chose not to continue interventions that were boring, worsened pain and could lower their mood.

> Adolescent 11: I had to do this whole thing of I had to sit there and look at it [my leg] in the mirror, I couldn't do that, I couldn't sit there and look at something I

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> hated. I found myself blocking myself even more from it because I didn't like it. And even now if you ask me to do that my mood will completely change, I stop, don't want to talk, I don't want to do anything.

Any intervention, if specifically linked to a meaningful participation in life, allowed young people to visualise how interventions could be part of their path to recovery. Adolescents described positive effects that included increased function, a sense of social belonging, happiness and reduced pain.

> Adolescent 21: We did the mountain, set goals, it just made me like quite happy talking about the things that I can be doing soon... just making it fun.

One of the narratives described how vital the change from a biomedical to a person-centred approach was to their recovery. When receiving outpatient physiotherapy, the young person described being so fearful of physiotherapy sessions that they had panic attacks prior to each appointment. In comparison, when they were admitted to the hospital for intensive physiotherapy, despite having the same physiotherapist, they reported dramatic improvements that led to a full recovery. This quote was their explanation of why they believed physiotherapy intervention was effective as an inpatient.

> Adolescent 3: I think it was more kind of personal. Because you're there all the time they're seeing you every day twice a day it was more kind of, friendly... It's less kind of you get there, you do it and then you go...I think the less exercise-based stuff and more the kind of just walking and. It's not just focused on the foot.

These findings question whether physiotherapy outpatient departments may historically, through their timings and physical layouts, facilitate a biomedical approach that views people through their problems and not as a person. The flexibility and time allocated to inpatient and tertiary services appeared to facilitate a holistic view more readily, but this was not always the case. In outpatient settings, when young people were taken out of the clinic environment and were in a hydrotherapy pool, gym or walking outside, this same wider focus on the individual was described.

4 | Discussion

To the best of our knowledge, this is the first study to explore adolescent and parent perceived effectiveness of physical interventions for chronic musculoskeletal pain within usual healthcare. The real-world setting allowed a diverse range of physical interventions, delivered across inpatient and outpatient services, to be explored and common themes established. The themes convey the importance of therapeutic alliance, intervention context and the need for personalised care. Themes link to the self-determination theory, where basic psychological needs for autonomy (self-endorsement of behaviour and sense of willingness), competence (experience of behaviour as effectively enacted) and relatedness (sense of being cared for and connected with others) affect the type and strength of motivation (Deci and Ryan 2008; Niemiec and Ryan 2009; Ryan and Deci 2008).

Having a choice and the opportunity to play an active role in decisions, treatment and goals, appears to influence adolescent experience and perceived effectiveness of physical interventions. Riggenbach et al. (2021) and Palermo et al. (2007) found that adolescent autonomy is related to levels of function (vs. activity avoidance) in those affected by chronic pain. In qualitative studies, adolescents reported that physiotherapy provided the opportunity to regain some control over their body and life and the confidence to influence their own situation and self-manage musculoskeletal conditions (Ahlqwist and Sällfors 2012; Blanco-Morales et al. 2020; Williams et al. 2015). The transition from being autonomous during physiotherapy, to taking the lead and self-managing musculoskeletal pain is an area that requires further research (Paskins et al. 2022). Straszek et al. (2023) found that adolescents reported multiple barriers to self-management of back pain and sought guidance and validation from health professionals. Contrary to adult populations, a triadic relationship (adolescent, parent and physiotherapist) exists in adolescent care (Miciak 2020). Physiotherapists need to acknowledge the known tensions between the duty of parents to protect adolescents as well as respect and enhance their autonomy (Brierley and Larcher 2016).

Therapeutic alliance, described within a psychotherapy context, as an affective bond between the patient and therapist and agreement on goals and tasks (Bordin 1979) appeared to be a key factor influencing perceived effectiveness. In adult literature, a strong therapeutic alliance appears to be more effective than traditional physical interventions alone for the treatment of chronic musculoskeletal pain (Kinney et al. 2020). Specifically, in the field of paediatric physiotherapy, Crom et al. (2020) explored therapeutic alliance with children receiving physiotherapy due to a range of neurological, orthopaedic and congenital diagnoses. Crom et al. (2020) acknowledged the differing roles and responsibilities of parents, physiotherapists and young people and the complexities of a triadic therapeutic alliance. Trust was a key concept that appeared to be a prerequisite to treatment. An area not considered in previous research but identified in the current study was the importance of mutual agreement on the cause of pain. The need for credible explanations has been identified in this population (Djurtoft et al. 2024; Joslin et al. 2024) and may present a unique component in developing therapeutic alliances that warrant further exploration.

Adolescents and parents expressed that it was how the intervention was delivered, who delivered it and where it was delivered that mattered. Understanding the impact of these contextual factors appeared vital to explaining why physical interventions were considered effective from adolescent and parent perspectives. Contextual factors are defined as components of all therapeutic encounters that 'may constitute the entirety of the perceived effects of the intervention itself or be additive to effects of the interventions' (Cook et al. 2023, 4). Contextual factors have been poorly conceptualised and underreported in paediatric literature (Levac 2024), with current definitions on the basis of adult evidence (Cook et al. 2023; Weinbrecht-Mischkewitz

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et al. 2024). The current study found that interventions that focused on a young person's participation in life were perceived as more effective than interventions that focused on bodily impairment. When physical interventions linked to the real-world and a meaningful goal, treatment became enjoyable. The scoping review by Holt et al. (2020) found exercises that were fun or modified to include enjoyable activities that facilitated exercise adherence for children and adolescents affected by musculoskeletal conditions. Physical interventions appeared more effective when young people performed a physical intervention in the presence of a trusted physiotherapist in a new environment. The importance of having physical interventions supervised by a physiotherapist was supported by Ahlqwist and Sällfors (2012) who found that adolescents felt able to expose themselves to situations they had previously avoided when physiotherapists were present. Subtle differences in the delivery of physical intervention could change adolescents' expectations of their ability to perform the activity and could lead to a new expectation of recovery. Conceptual frameworks, such as the ecological-enactive perspective (Vaz et al. 2023), which combine this dynamic interplay between personal and environmental contextual factors could offer guidance to physiotherapists on how to help adolescents perceive and enact opportunities for action on the basis of their goals.

Study limitations should be considered when interpreting the findings. Firstly, it is important to acknowledge that the study involved secondary analysis, and therefore, the interview guide was not developed to directly answer the research question of this study. Notably, this study only explored the perspectives of adolescents and parents, and not the experiences of health professionals. This warrants further exploration to establish a complete picture of perceptions. Additionally, the current study context was experiences within NHS outpatient and inpatient services in England. Relatedly, it would be beneficial to conduct similar studies in different countries and healthcare settings where paediatric services differ. In terms of participant characteristics, not collecting sample characteristics of race, ethnicity and socioeconomic status hindered the understanding of sample diversity and representativeness and would need to be captured and described in future work.

5 | Conclusions

There were no ideal physical interventions: individuals had their own story, their own perspective of what worked and this changed in real-time during interactions inside and outside of physiotherapy sessions. Core concepts that appeared to influence the perceived effectiveness of physical interventions, from the perspectives of adolescents and parents, included facilitating adolescent autonomy, establishing a therapeutic alliance based on trust and focusing physiotherapy on meaningful participation in life. Social–contextual factors may provide physiotherapists with insights into why satisfying basic psychological needs of autonomy, competence and relatedness may impact behavioural change and the perceived effectiveness of treatment. Future research should explore how to personalise physical interventions so that adolescents, supported by their parents and physiotherapists, can gain confidence in engaging in activities they enjoy. Further research to understand the mechanisms of change during physiotherapy treatment can inform clinical practice recommendations and ultimately improve health outcomes in this under-served population.

Author Contributions

Rhiannon Joslin: conceptualization, methodology, analysis, writing–initial draft, writing–review and editing. **Lisa Roberts:** conceptualization, supervision, analysis, writing–review and editing.

Acknowledgements

We would like to acknowledge the contribution of Dr Maggie Donovan-Hall in the primary programme of research.

Ethics Statement

Ethical HRA approval was granted (IRAS ID: 230317 REC Reference number: 18/SC/0138).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The qualitative research raw data will not be shared.

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